

PLATO RISING

Online learning for Atarians

by DAVID & SANDY SMALL

Atari's PLATO cartridge (officially known as the Atari Access Software for the PLATO Services Network) is the Atari user's link to Control Data Corporation's PLATO network, the "ultimate" electronic information and communication utility.

PLATO was developed in 1962 at the University of Illinois. After 14 years of development, Control Data (CDC) bought the marketing and trademark rights to the PLATO software.

In the early 1970's, a number of large corporations began to develop specialized vocational software for use on CDC's main-frame computers. This software was used to tutor employees, who accessed the material through expensive remote graphics terminals. There was a \$50/hour surcharge for use of CDC's computers, so only enormous organizations such as Boeing, Lockheed and major universities could afford to use them.

Then, during the mid-1970's, CDC began to sell Cyber supercomputers running the PLATO operating system to corporations and universities all over the world. Electronic mail, interactive games, bulletin boards, and many other services associated with electronic utilities evolved on PLATO during this period.

By 1978, the PLATO terminal evolved into a high-resolution, touch-sensitive screen with a

dedicated keyboard, graphics printer and 1200-baud modem. As personal computers became more capable, CDC developed terminal "emulators" that gave some PC's, such as the Zenith Z-100 and the IBM PC, the ability to connect to the PLATO system. This eventually led to the Homelink "after-hours" service, which was affordably priced at \$5 per hour. As a result, a whole new generation of users discovered PLATO.

According to officials at Atari Inc., Atari's access software for the PLATO Homelink service should be released by the third quarter of this year. For stylistic reasons, we'll refer to this product as the Atari PLATO cartridge, although this is not its official name. —ANTIC ED

If you're an Atari computer user, you should be using PLATO.

Does this statement strike you as being too broad, too dogmatic? Well, it isn't — it's simply the truth. With its vast variety of applications and uses, PLATO has something important to offer every Atari user.

Are you interested in educational software? More than 200,000 user hours worth of courseware — the largest educational-software base in existence — is available on this extraordinary system. This software is much more sophisticated than conventional text-only software — in fact, its combination of

continued on page 38





high-resolution graphics, informative text and user-friendly help features make it the best instructional system currently available. All in all, we have no hesitation in calling PLATO one of the finest communications systems ever devised.

PLATO began as an instructional computing system on an enormous Control Data Corporation (CDC) mainframe computer more than 15 years ago. Since then, PLATO has evolved into a remarkable communications system that links thousands of users across the North American continent and around the world. As a result, an international community — a world community, the first of its kind — has developed.

Thousands of people use PLATO every day to send mail ("Personal Notes") to each other, to communicate via bulletin boards or to do coursework. An entire PLATO-based society has sprung up. Marshall McLuhan's "global village" has become a reality.

PLATO's central system is located in Minneapolis, Minnesota, the home of CDC. Telephone and satellite links connect it to systems at the University of Delaware, Florida State University, and the University of Illinois, as well as to major systems in Australia and Belgium.

THE ATARI PLATO CARTRIDGE

Until recently, however, access to PLATO was restricted to those who could afford the cost of a dedicated PLATO terminal — \$1000 per month — or people who used it at work or school. After using PLATO in college on an experimental basis, it was difficult for us to give it up. Interaction with other users, "notes-files" that provide a medium for stimulating discussions and arguments every day, battles in the depths of ancient dungeons or in the far reaches of outer space — all of these were lost to us. So imagine our excitement when we heard that Atari would soon be offering a cartridge that could access PLATO!

The **Atari PLATO** cartridge allows Atari computers to emulate a standard PLATO terminal, receive and send text, display graphics, and use PLATO's special keyboard commands. Some limitations are involved if you use this kind of hookup, as compared to using an actual PLATO terminal, but they're insig-

nificant when you consider the low cost of an Atari computer and the awesome power of the PLATO system.

GRAPHICS

One of Atari PLATO's limitations involves its graphics-display capabilities. It uses a display made up of a 512×512 grid of black-and-white dots; these are used to form both letters and graphics characters. This display is similar to the Atari's highest resolution mode, Graphics 8 (or ANTIC mode F) — which has a resolution of 320×192 — but has been "squeezed" to fit into the Atari screen. This works quite well, except when very detailed graphics are involved. And it's even possible to view a portion of the 512×512 screen at full resolution (thanks to the efforts of Atari designer Vincent Wu) when this is necessary for detailed work.

GENERATING SPECIAL COMMANDS

To duplicate the standard PLATO keyboard, the Atari computer must generate a number of special keystrokes. These are used to control your progress through a program, or, in PLATO terminology, a "lesson." Examples include "DATA," "LAB," "NEXT," and "HELP." To generate these commands, simply press the [START] button and the first letter of the key's name. For example, to "press" the DATA key, hold down [START] and press "d." Within fifteen minutes or so, you should be familiar and comfortable with these commands and the special requirements of the Atari PLATO cartridge.

USING THE CARTRIDGE

To use Atari PLATO, you need the Atari PLATO cartridge, an Atari 850 interface and a modem, which can be either a 300-baud or a 1200-baud model. The modem need not be the model manufactured by Atari, but a 1200-baud unit is strongly recommended. These are four times faster than their 300-baud brothers, and this makes a big difference on PLATO. A tremendous amount of information on the system is available, and the faster you can get it, the better.

The interface and modem are used to dial PLATO's local number in your com-

munity. One of PLATO's major advantages is that you can access it by means of a local telephone call — you don't have to worry about calling Minneapolis long-distance.

CONNECTING WITH PLATO

When you first dial up PLATO with your new cartridge, there are a number of initial steps you must go through. At first they'll be confusing, but with a little practice you'll breeze through the procedure. As with every other aspect of PLATO, the log-in procedure has been tested by tens of thousands of users and has been made as simple and straightforward as possible.

Essentially, CDC provides you with a local dialup number (for 300- or 1200-baud operation) and a "signon" code that consists of your "name" and "group." These are abbreviated as follows: "name / group." Get used to this syntax; you'll be known by your "name / group" no matter what you do on PLATO.

You will also be given a system name; a full signon consists of the following elements: "name / group / system." Since you can get mail from people on other systems, or communicate with them by means of "Notesfiles," you'll soon run across the names of some of the systems other than your own.

Local log-in procedures vary to a certain extent; consult your PLATO cartridge and Control Data Homelink manuals for details.

TERM-TALK

Between ten and 600 people are likely to be logged onto PLATO at any given time, and many of them will show up on the user's list — the others are engaged in work that can't be interrupted. Press "user list" and you'll see the name/group of each logged-in user.

You can talk to any PLATO user directly via "TERM-talk." TERM is a special PLATO key. To initiate this means of communication, press START-t. When the message "what term?" appears at the bottom of the screen, type "talk" and press [RETURN]. You will then be asked for the name and group of the person you want to talk to. Once you've provided this information, PLATO will "ring" the other person.

VINCENT WU,

Designer of Atari's PLATO Cartridge

Vincent Wu, designer of Atari's "Access Software for the PLATO Services Network," has accomplished something many said couldn't be done — he's made the powerful PLATO educational service run on Atari computers.

Born in China in 1942, Vince emigrated to the U.S. at age nine, and lived in New York and San Francisco before acquiring an A.B. in physics from San Jose State University in 1964. He earned a Master's Degree in Mathematics in 1966 at the University of Illinois, Urbana, where he then worked with Paul Tenczar, the originator of the PLATO system language called TUTOR, at the Computer Educational Research Laboratory (CERL).

At the U of I Medical Computer Lab, Vince was involved in the development in 1976 of the first minicomputer base for the PLATO educational operating system.

The PLATO project at Atari was started in 1981 by Lane Winner. He wrote a working PLATO emulator (Version 1) that was strictly 300 baud, and displayed only one segment of the PLATO screen at any given time. Vince then worked with Joe Miller on Version 2 of the terminal-emulator program, which was presented to Control Data Corporation (CDC) in December of 1981. According to Wu, this was the first time a microcomputer was used to access the CDC PLATO system.

By early 1983, however, negotiations



LINDA TAPSCOTT

between CDC and Atari had broken down, and the PLATO emulator project at Atari was — temporarily, it turned out — dead. Due to Vince's concerted efforts, which included many hours of work on his own time, Atari's top management decided to give the project another chance. As a result, Vince was the sole designer of Version 3 of Atari's PLATO access software, and was instrumental in initiating the negotiations that brought CDC and Atari back together on the project.

Vince was the first designer to write an algorithm that could compress the 512×512 line PLATO screen into the 320×192 lines found in Atari home computers (Graphics Mode 8). Because this compression resulted in the loss of some fine detail, he added a zoom feature that magnifies chosen areas of the display. He also emulated PLATO's touch screen with a joystick-controlled cursor. Because of these and other features, an Atari with Vince's cartridge installed can run four times as much of PLATO's courseware as the IBM PC.

Vince's cartridge for PLATO is the only Atari product that displays 64 columns by 32 lines of text on a normal TV screen. It also allows the user to adjust the color and brightness of both image and background. In addition, it can be used with a wider variety of modems than any other Atari product.

system; just type "TERM-consult" and a consultant will be on-line in a few moments. Don't be afraid to call a consultant — they're there to help. We've been helped by "Chip" Aspnes, Barry Doolittle, Chuck Miller, Scott Rautmann, and Chris Johnson, among

others. You'll be glad you got to know them.

LEARNING BY WATCHING

PLATO also offers a special monitor mode that lets you learn by watching others learn. It allows you to "monitor" the progress of another user (with his or her permission, of course!). You see the same things on your screen that the other user sees, and the two of you can communicate with each other during the monitoring process. Some of PLATO's complex games, such as Moria or Empire, are best learned through use of the monitor mode.

NOTESFILES

Notesfiles are a much improved version of computer bulletin boards such as those available on **CompuServe** or **The Source**, or on public-domain bulletin board systems (BBS's). Such boards usually allow users to talk to each other about a particular subject. PLATO's notesfiles, however, have an added advantage: ten years of use, and the useful feedback provided by numerous users over that period of time.

Let's look at a sample notesfile — in this case, one that is devoted to discussion of Atari computers. To access it, select the "enter a notesfile" option and type in the name "atari." Next, you'll see a box in the upper-left-hand corner of the screen. This box lists the titles of the notes in the file and the number of responses to each of them. Notes are numbered from "1" through "??," up to 99 responses can be made to each note.

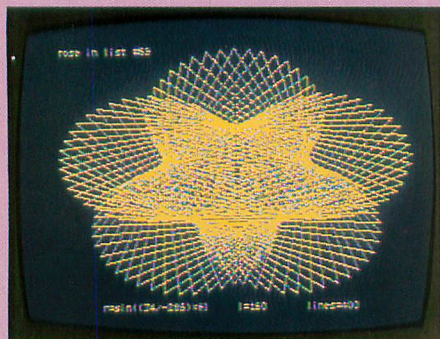
It's always a good idea to read the "policy statement" for a notesfile that's new to you before you start to use it, so you can make sure you understand what the file is about and if you should use it. You should use a notesfile only to discuss the subject it was designed to cover.

For instance, you can talk about Atari machines and related topics in =atari=. Microcomputers of all kinds are discussed in =micronotes=. Political discussions are held in =forum=, and =pad= is the place for offbeat humor.

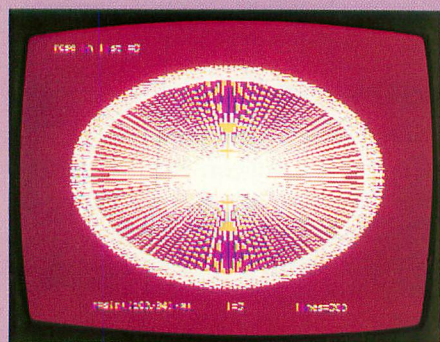
When you have a chance, read the =atari= policy note. Surprise: David left it! It includes the date and time he

continued on next page

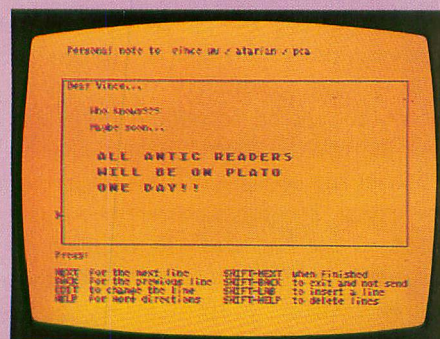
TERM-talk gives you the opportunity to learn about the PLATO system from other users, who are usually happy to share suggestions about lessons and notesfiles. In addition, special consultants are available to answer questions or solve problems concerning the



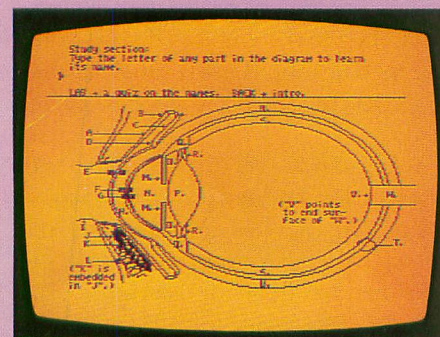
This illustration of a graphing equation is taken from PLATO's "Rose" (=0rose=) program.



Another sample of a graphing equation, this screen shot illustrates the user's ability to alter background and foreground color and luminosity.



PLATO's personal note (email) program allows you to send notes to users on other (linked) computers.



In "Eyeq" (=0eyeq=), graphics and animation are used to reveal the eye's anatomy.

left the note, along with his signon, and is an example of a typical note on PLATO, whether it's a personal note (or mail, which is designated for one person) or a group note (or a notesfile, open to anyone who wants to read it). Each note can be a maximum of one page in length.

ACCESSING NOTES AND RESPONSES

After reading the =atari= policy note, press [RETURN] and you'll be back at the =atari= notesfile page. Pick a note that looks interesting and type in its identifying number. For instance, a note's title might be "850 Interface" and its number 134. To access it, type in 134 on the "What Note?" line. The note will then appear onscreen, along with the date and time it was left and the name of its author.

If there are any responses to the note, you'll be prompted to "Press LAB to see response." You access the responses, one at a time, through successive presses of LAB. These responses are not the main notes, and do not appear on the notesfile index page. They are simply responses to the "base" note. If you want to leave a comment about a base note, leave it as a response; if you'd rather start a whole new note, leave a new base note instead.

There are currently more than 100 notes in =atari=, some of them going back more than two years. Hundreds of responses are also listed, and =atari= is not even a particularly active notesfile! The busier files, such as =micro=, are winnowed down on a daily basis — they receive hundreds of notes and responses every day.

A LIST OF POPULAR NOTESFILES

Some of the more popular notesfiles are listed below. Bear in mind, however, that there are more than 10,000 notesfiles currently active on PLATO (most of them dedicated to very specialized purposes); this selection is intended to cover only a few of the most widely used files.

- =forum= . If you have a yen to discuss politics or current events, =forum= is for you. It includes input and opinions from all points on the

political spectrum, so feel free to jump up on your soapbox and spout off.

- =micronotes= . An extremely busy file that consists of news and discussion of microcomputers of all kinds. Questions are most welcome. Currently, Apple's new Macintosh computer is a hot topic, and six pages of Mac specifications are available, but there's also plenty of news about Atari, IBM, and other micros and microprocessors.

- =pad= . A great place to let loose. Look into it for yourself; it's impossible to describe.

- =parents= . The place to ask questions — and get answers — about raising children. Right now, someone wants to know when to let their teenage daughter use makeup.

- =tavern= . A notesfile dedicated to "labyrinth," a fantastic underground-dungeon game.

- =starbase= . This file is used by the Federation Team in the interstellar game =empire= . Other notesfiles are used by the game's other teams, the Kazars (formerly the Klingons), the Romulans, and the Orions. New members are always welcome; if you're interested, leave a message at the door.

- =womannotes= . A place to discuss women's issues. Right now, the ERA (Equal Rights Amendment) is the main subject under consideration.

PERSONAL NOTES

To move out of notesfiles and into Personal Notes, press Shift-BACK several times and then select the Personal Notes option. As a new user, you won't have any incoming mail (or "pnotes") to contend with right away, but active PLATO users receive and generate all sorts of mail. By the way, a pnote cannot be read by anyone — including the PLATO system's staff — other than its intended recipient. Your privacy is closely guarded by the system and its staff.

As a PLATO user, you're likely to spend a great deal of time in Personal Notes, because they're a great way to communicate with other users. Again, like all notes on the PLATO system, pnotes can be a maximum of one page in length. The system automatically lets you know when you've received a pnote.

ASKING FOR HELP

When using the PLATO system, you can ask for help at any time by pressing the HELP key (Start & H). On-line help will be made available immediately, and all the options open to you in a given situation will be explained. Don't be afraid to follow the instructions you are given — much of the material in this article is available through HELP, and, in fact, it was the source of much of the material that appears in this article.

In addition, lessons about using Personal Notes, Notesfiles and TERM-talk are offered in PLATO's voluminous library, which is listed in the main menu. PLATO is more than happy to teach you everything you need to use it.

PLATO'S GAMES

Let's take a look at the selection of games available on PLATO. After all, PLATO was a great source of games — many of them involving more than one player — long before the Atari even existed. You can compete against, or work with, other users from around the U.S. or around the world. What a great way to play! The only thing I know of that compares to it for the Atari is **COMM-BAT**, a two-person game from Adventure International, and even COMM-BAT doesn't really compare to the classic PLATO games.

An entire section of the PLATO library is devoted to games. It includes all the games you could possibly want. Look them up — the following are just a few of the highlights:

- =moria= and =labyrinth=. These are classic dungeon games in the Dungeons and Dragons mold. You can play them by yourself or with a group of other users. They offer you the chance to explore multiple-layer dungeons filled with deadly traps, monsters galore and great treasure. Along the way, you accumulate experience, wealth and magic weapons. You can also help others in your group if (when!) they run into trouble.

- =drygulch=. This mining game features a mine and ghost town with supernatural characteristics.

- =empire=. The space game — it has no equal. As many as 32 players, each working a separate terminal, can

command spaceships involved in a battle for control of the galaxy. Four teams struggle day in and day out, every second the system is up, to conquer the universe. To do so, your team must take over the other teams' home systems along with a number of independent planets, but — remember — at the same time the other teams are trying to do the same thing to you.

A help lesson for =empire=, complete with tips for beginners, is available on PLATO; be sure not to miss it. And get ready for some of the fiercest, fastest gameplay you've ever encountered.

We should mention that one of the designers of =empire= is Chuck Miller ("chuck miller / pso"), one of the system consultants mentioned earlier. If you like =empire=, drop him a note and tell him. He's devoted a great many hours to making =empire= the best possible game of its kind.

- =0airfight=. This dogfight game features 3-D graphics, and is superior in many ways to any of the flight-simulator games on the market.

ADMISSION TO A NEW SOCIETY

We've been trying throughout this article to emphasize the human side of PLATO. Many people have met on PLATO, and quite a few romances, marriages, and lasting friendships have resulted. We can't even begin to count the number of friends we've made on the system. PLATO is *about* people — it's not just an educational network. In effect, for a very low admission charge, PLATO offers you the chance to join the world's first electronic society.

We've made friends with fellow PLATO users from places as diverse and distant as England, Sweden, Venezuela, Belgium, Australia and Alaska. This new society is scattered across the face of the globe, and it gives you the opportunity to meet an incredible number and variety of people.

The Atari PLATO cartridge itself is only \$49.95, and connect time on the system is apparently going to cost about \$5 an hour. This puts PLATO in the same league with CompuServe and The Source in terms of cost, but in other respects there's no comparison. Compu-

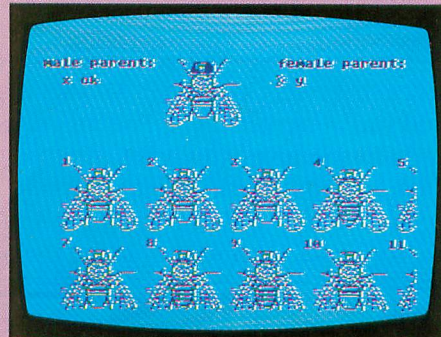
continued on page 42



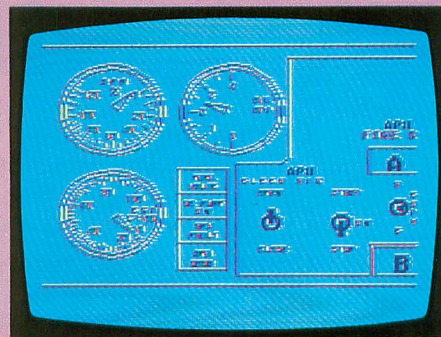
"Lamaze" (=0lamaze=) is a tutorial that prepares women and their partners for childbirth.



Players all over the country wander through the 248 mazes of "Moria" (=0moria=), PLATO's most popular dungeon game.



"Fly" (=fly=) offers an interactive course in genetics that includes the cross-breeding of fruit flies.



The "apu747" (=0apu747=) simulation by Boeing Aircraft teaches ground crews how to start a jumbo jet with an auxiliary-power-unit truck.

Serve is just beginning to explore the potential of this kind of communications network; PLATO has known about it for ten years or more. We think you'll find PLATO to be a superior system, especially in terms of its educational applications.

SAVING THE BEST FOR LAST

PLATO began as an educational system, and, indeed, as an experiment to determine how computer-based education could be most effectively explored. Because of these roots, PLATO places a heavy emphasis on computer education. In fact, that's all CDC wants to talk about. Perhaps they haven't yet realized that the communications aspects of the system are a potential gold mine for them. If the Atari community hooks up to PLATO in force, though, CDC may finally see the light.

At any rate, as we mentioned earlier, there are more than 200,000 hours of courseware available on PLATO, which means that you'd literally have to spend 200,000 hours in front of your Atari to go through the entire PLATO library. Courses on every imaginable subject are included.

Would you like to learn about sentence structure? Go to the PLATO library and run =0sentences=. This amazing little lesson gives you a list of words such as "boy," "girl," "dog," "carries," "over," "under," and so on, and asks you to construct a sentence out of them

(e.g., "The boy carries the dog over the car to the girl."). In a few seconds, an animated sequence appears on the screen, showing a boy carrying a dog over a car to a girl. This is an extremely powerful teaching tool, and quite a programming feat as well.

Having trouble with arithmetic? PLATO abounds with lessons on everything from the fundamentals of math to the mysteries of calculus. Interested in chemistry? PLATO can show you how to set up a distillation apparatus for a sample experiment, explain how to proceed with the experiment itself, and even show you the end result — an imaginary explosion — if you let the solutions get too hot.

PLATO's role is to help you learn, not to make you fail. Its lessons are not designed with failure in mind. If you're having trouble, you can press HELP (or be taken to help automatically) as often as necessary until you understand the lesson. PLATO is the epitome of individualized education — it's like having a powerful and extremely patient teacher who's been assigned to work with you on an individual basis. No deadlines are involved, and there is no peer pressure to contend with.

A FINAL NOTE

We'd like to take this opportunity to thank Vincent Wu, the designer of the Atari PLATO cartridge, Atari Inc. and Control Data Corporation for putting PLATO within virtually everyone's

reach. As Atari users and observers for four years now, we're convinced that the introduction of the PLATO cartridge is the most significant event in the history of the Atari home computers.

This one step has given the Atari access to many megabytes of memory, thousands of man-years of software and experience, the incredible computational power of a Cyber 800 mainframe computer, and the best communications network in existence. The Atari has become a supercomputer! But PLATO's most powerful selling point is its element of humanity — its ability to put you in touch with all those other users out there. This feature should make it very seductive to Atari computerists, many of whom are used to working by themselves on a machine day after day, or night after night. Suddenly, all of you, you're no longer alone.

David and Sandy Small are professional programmers who work extensively with Atari computers and Atari-compatible peripherals to create software for the Atari. They've spent several months testing a prototype of the Atari PLATO cartridge, which is scheduled for release later this year.

Antic is planning to maintain an =antic= notesfile on PLATO. It will be open to all users who want to leave questions or comments. To sign on, type "b" for electronic mail, "n" for notesfiles, and "antic." —ANTIC ED 

now

ONLY \$49.95!

USE MODEMS & RS-232 PERIPHERALS DIRECTLY WITH YOUR ATARI!

R-VerterTM* SERIAL BUS MODEM ADAPTOR

Compatible with —

- Anchor Automation (including VolksmodemTM — no cable required)
- Hayes • Novation • and Others

Comes with Software —

Including Smart Terminal Emulator (with X Modem) and "R:" Handler

Works with — Atari 400TM; 600XLTM; 800TM; 800XLTM**

Also Offering — INTERFAST-ITM*

Buffered Printer Interface —

- 4K Buffering - Programmable - Print Custom Character Sets • Easy to use
- Compatible with all Software

ONLY \$169.95!

AID
ADVANCED INTERFACE DEVICES

* Trademarks of Advanced Interface Devices, Inc.
** Trademarks of Atari, Inc.

P. O. Box 2188
Melbourne, FL 32902
(305) 676-1275